

Isolation Forest — A One-Page Primer

For Task 5 of the Askari Bank AI/ML Internship Assessment

What it is

Isolation Forest is an unsupervised anomaly detection algorithm (Liu, Ting, Zhou; 2008). It's tree-based, fast, and widely used as a baseline for detecting outliers in numeric data. Implemented in scikit-learn as `sklearn.ensemble.IsolationForest`.

The core intuition

Anomalies are **few and different**. If you build random binary trees that split the data at random features and random split values, anomalies get isolated (separated into their own leaf) with fewer splits than normal points — because they sit in sparse regions of feature space. The algorithm averages this isolation depth across many trees. Shallow average depth → anomaly. Deep average depth → normal.

Basic usage

```
from sklearn.ensemble import IsolationForest
import pandas as pd

# X must be a 2D array-like of NUMERIC features only.
# You'll need to engineer numeric features from raw transaction data.
X = df[numeric_feature_columns]

model = IsolationForest(
    n_estimators=100,
    contamination=0.05,  # expected fraction of anomalies – must guess
    random_state=42,
)
model.fit(X)

df['anomaly'] = model.predict(X)  # -1 = anomaly, 1 = normal
df['score'] = model.score_samples(X) # lower = more anomalous
```

The contamination parameter — read this carefully

contamination is the expected proportion of anomalies in your data. You have to choose it. Too high → normal points get flagged as anomalies (false positives). Too low → real anomalies go undetected (false negatives). The default (0.1 in older sklearn, 'auto' in newer) is rarely the right choice. For banking transaction monitoring, real-world contamination is usually 1–2% or lower. Part of this task is picking a defensible value and justifying it.

When Isolation Forest works well

Numeric features, many of them. Anomalies that are genuinely "different" in a statistical sense (unusually large or small values across one or more dimensions). A reasonable prior on the contamination rate.

When it struggles

Mostly categorical features — you have to encode them (one-hot, target encoding, etc.), which often loses structure. Contextual anomalies — patterns that are normal in one context and anomalous in another (e.g., a large transaction is normal for a business account but abnormal for a pensioner's account). Isolation Forest has no concept of customer-level context unless you engineer it into the features yourself. Very small datasets where trees don't have enough to split on.

Hint — not a solution

The transactions dataset has categorical columns (txn_type, channel, city) and numeric columns (amount_pkr, plus whatever you derive — hour of day, day of week, customer aggregates). You decide which features to include, how to encode categoricals, and how to handle the fact that a PKR 50,000 withdrawal means different

things for different customers. Defend your choices in your answer.

Further reading (not required)

sklearn docs: scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html

Original paper: Liu, Ting, Zhou (2008), "Isolation Forest", ICDM.